



B.M.S. COLLEGE OF ENGINEERING, BENGALURU-19

Autonomous Institute, Affiliated to VTU

DEPARTMENT OF MATHEMATICS

Semester & Branch:		Fourth Semester CS cluster except AIML & CSBS		Subject:	Linear Algebra and Optimization	Course Code:	23MA4BSLAO	
Time:	12:45 PM – 2:00 PM		Test Date:	24/04/2025		Max. Marks:	40	
Test No.	Q. No.	1. Questions from 1 to 5 are compulsory 2. Answer either Question 6a or 6b					Marks	CO
Test-2	1	Given $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$. Find A^8 using Cayley-Hamilton theorem with minimal conventional operations.					6	1
	2	Let $\Delta(t) = (t - 2)^4(t + 5)^3$ and $m(t) = (t - 2)^2(t + 5)^3$ be the characteristic and minimal polynomial of T respectively. Obtain all possible Jordan canonical forms of T .					6	1
	3	Given $M = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. Find the algebraic and geometric multiplicity of each eigenvalue λ of M and hence find Eigen space corresponding to smallest eigenvalue of $M^2 + \frac{1}{2}M$.					7	1
	4	Find the characteristic and minimal polynomial of the matrix $B = \begin{pmatrix} 2 & 5 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 2 & 0 \\ 0 & 0 & 3 & 5 & 0 \\ 0 & 0 & 0 & 0 & 7 \end{pmatrix}$.					7	1
	5	Find the modal matrix P which diagonalizes matrix $A = \begin{bmatrix} 0 & 0 & -2 \\ 0 & -2 & 0 \\ -2 & 0 & 3 \end{bmatrix}$.					7	1
	6a	Orthogonally diagonalize the matrix $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ and hence find A^{-4} .					7	1
	OR							
	6b	Let $T: S \rightarrow S$ be the linear transformation defined by $T(A) = BA + AB$, where $A \in S$, and $S = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} : a, b, c \in \mathbb{R} \right\}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Find the Eigen space corresponding to largest eigenvalue of T .					7	1
Course Outcome:								
CO1	Apply the concepts of linear algebra in Computer and Allied Engineering Sciences.							